```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

volatile sig_atomic_t stopped = 0;

void handler(int sig) {
    stopped = 1;
}

int main() {
    struct sigaction sa = {0};
    sa.sa_handler = handler;
    sigemptyset(&sa.sa_mask);

    if (sigaction(SIGTSTP, &sa, NULL) == -1) {
        perror("sigaction");
        return 1;
    }

    printf("PID: %d\n", getpid());
    printf("Press Ctrl+Z to request suspension.\n");

    while (!stopped)
        pause();

    printf("SIGTSTP received. Resuming in 3 seconds...\n");
    sleep(3);
    stopped = 0;

    printf("Program continuing.\n");
    while (1)
        pause();

    return 0;
}
```

```
  GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <signal.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

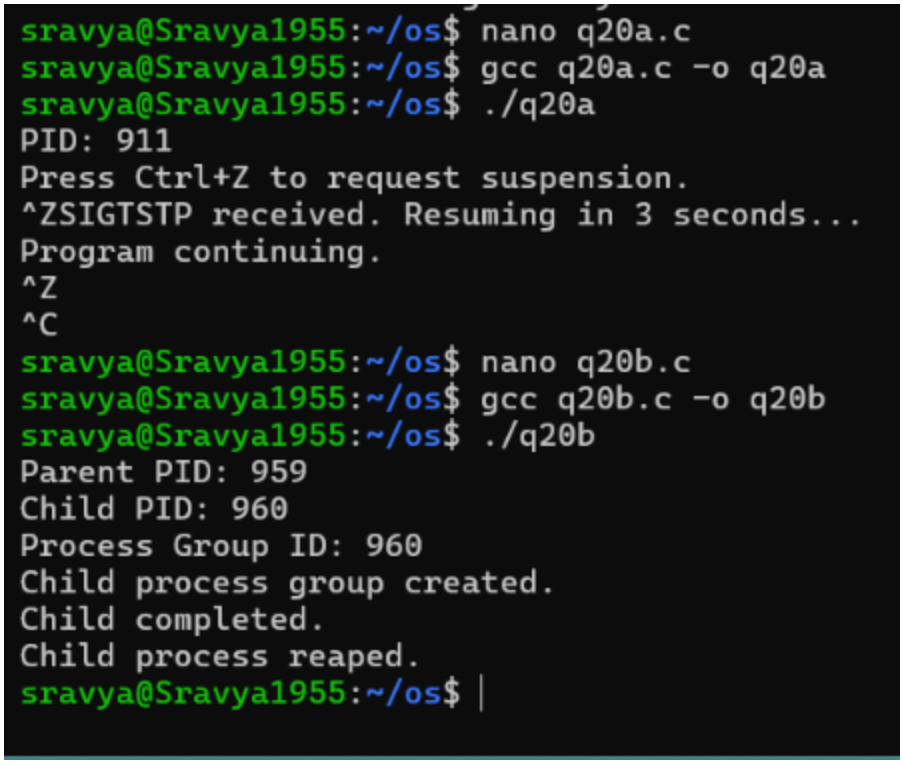
    if (pid == 0) {
        setpgid(0, 0);
        printf("Child PID: %d\n", getpid());
        printf("Process Group ID: %d\n", getpgrp());
        sleep(3);
        printf("Child completed.\n");
        return 0;
    }

    setpgid(pid, pid);

    printf("Parent PID: %d\n", getpid());
    printf("Child process group created.\n");

    waitpid(pid, NULL, 0);
    printf("Child process reaped.\n");

    return 0;
}
```


sravya@Sravya1955:~/os$ nano q20a.c
sravya@Sravya1955:~/os$ gcc q20a.c -o q20a
sravya@Sravya1955:~/os$ ./q20a
PID: 911
Press Ctrl+Z to request suspension.
^ZSIGTSTP received. Resuming in 3 seconds...
Program continuing.
^Z
^C
sravya@Sravya1955:~/os$ nano q20b.c
sravya@Sravya1955:~/os$ gcc q20b.c -o q20b
sravya@Sravya1955:~/os$ ./q20b
Parent PID: 959
Child PID: 960
Process Group ID: 960
Child process group created.
Child completed.
Child process reaped.
sravya@Sravya1955:~/os$